

ThinkersCad Vs Circuits Lab

Tinkercad vs. Your Simulator — Component Library Comparison

Method note: Tinkercad's list was pasted as raw sidebar/tooltip text with no section markup and some stray characters (numbers, arrows, part codes). I've grouped it by the same 11 section headers your simulator uses, and flagged any entry I couldn't confidently identify (e.g. SW 2000, 215, 711). Your library's list was parsed directly from the component-picker HTML, so those counts are exact.

1. Executive Summary

Section	Tinkercad (approx.)	Your Simulator (exact)	Shared	Tinkercad-only	Yours-only
General	6	10	6	0	4
Input	~22	35	~15	~5	~20
Output	~17 (unique types)	25	~14	~3	~11
Power	6	10	6	0	4
Breadboards	3	1	1	2	0
Microcontrollers	4	4	3	1	1
Instruments	4	4	4	0	0
Integrated Circuits	6	26	5	1	~20*
Power Control	~12	13	~8	~3	~5
Connectors	2	8	2	0	6
Logic	~22	26	~8 (conceptual)	~12	~15
Networking	0	3	0	0	3
Total	~104	~165			

- Most of your "Integrated Circuits" 74xx chips *do* exist in Tinkercad — just filed under **Logic**, not IC. See §3 "Cross-Section Differences."

2. Section-by-Section Breakdown

2.1 General

In both: Resistor, Capacitor, Electrolytic/Polarized Capacitor, Inductor, Diode, Zener Diode

Tinkercad has, you don't: (*none — full coverage*)

You have, Tinkercad doesn't:

- Schottky Diode
 - Fuse
 - Crystal Oscillator
 - Trimmer Capacitor
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2.2 Input

In both: Pushbutton, Rotary Potentiometer, Slideswitch, Photoresistor, Photodiode, Flex Sensor, Force Sensor, Ultrasonic Distance Sensor, PIR Sensor, Temperature Sensor, Soil Moisture Sensor, Tilt Sensor, Gas Sensor, Keypad 4×4, DIP Switch

Tinkercad has, you don't:

- Ambient Light Sensor (listed separately from Photoresistor)
- Generic "IR sensor" (distinct from your IR Receiver/Remote)
- DIP Switch **DPST** and **SPST ×4 / ×6** variants (you have one generic DIP Switch)
- **SW 2000** — unidentified part, likely a specific switch or sensor module

You have, Tinkercad doesn't:

- Water Level Sensor
 - Sound Sensor
 - Hall Effect Sensor
 - IR Receiver (as a discrete part, separate from a remote)
 - Real-Time Clock [DS1307]
 - Thermistor (NTC), as distinct from the general Temperature Sensor
 - Vibration Sensor
 - Magnetic Reed Switch
 - Phototransistor
 - Photo Interrupter
 - Toggle Switch
 - Slide Potentiometer
 - Trimmer Potentiometer
 - Rotary Encoder
 - Joystick
 - Limit Switch
 - Pushbutton (12mm) — size variant
 - IR Remote (*present in Tinkercad too, but filed under Output — see §3*)
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2.3 Output

In both: LED, RGB LED, Light Bulb, NeoPixel, NeoPixel Ring, NeoPixel Strip, Vibration Motor, DC Motor, Micro Servo, Hobby Gearmotor, Piezo, 7-Segment Display, LCD 16×2, LCD 16×2 (I²C)

Tinkercad has, you don't:

- DC Motor **with encoder**
- 7-Segment **Clock** Display
- Multiple pre-sized NeoPixel Ring/Strip presets (Ring-12, Ring-16, Strip-6/8/10/16/20) — you have one generic Ring and one generic Strip instead of fixed-size presets

You have, Tinkercad doesn't:

- LED (10mm) size variant
 - LED Bar Graph
 - LCD 20×4
 - LED Matrix 8×8
 - 4-Digit 7-Segment Display
 - Bi-Colour LED
 - Infrared LED
 - Continuous Rotation Servo
 - Stepper Motor
 - Solenoid
 - Speaker
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2.4 Power

In both: 9V Battery, 1.5V Battery, Coin Cell 3V Battery, Solar Panel/Cell, Potato Battery, Lemon Battery

Tinkercad has, you don't: *(none)*

You have, Tinkercad doesn't:

- VCC Supply (symbolic rail)
 - Ground (symbolic)
 - 1.5V Battery (AAA) — separate size from generic 1.5V cell
 - Battery Pack (4×AA)
-

2.5 Breadboards

In both: Breadboard

Tinkercad has, you don't:

- Breadboard Small
- Breadboard Mini

You have, Tinkercad doesn't: *(none)*

2.6 Microcontrollers

In both: Arduino Uno R3, ATtiny, micro:bit

Tinkercad has, you don't:

- micro:bit **with Breakout** board

You have, Tinkercad doesn't:

- Nano (ATmega328P)
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2.7 Instruments

Full parity. Multimeter, Power Supply, Function Generator, Oscilloscope — identical set on both sides.

2.8 Integrated Circuits

In both: 555 Timer, Op-Amp (Tinkercad: 741; yours: generic), Quad Comparator [LM339], Dual Comparator [LM393]

Tinkercad has, you don't:

- 556 **Dual** Timer

You have, Tinkercad doesn't (as an IC-section part):

- Shift Register [74HC595] / [74HC165]
- Decade Counter [CD4017]
- BCD → 7-Segment [CD4511]
- Darlington Array [ULN2003]
- ADC [MCP3008]
- EEPROM [24LC256]
- 3-to-8 Decoder [74HC138]
- 8-Channel Analog Mux [74HC4051]
- The whole 74xx logic family as *physical chips*
(74HC00/02/04/08/10/11/14/20/21/27/32/86/132)

⚠ Most of that last group **does exist in Tinkercad** — just as ideal parts in the **Logic** section rather than as chip-shaped ICs. This is a structural/categorization difference, not a true gap. See §3.

Optocoupler [4N35]: exists in both tools, but Tinkercad files it here (Integrated Circuits) while your simulator files it under Power Control.

2.9 Power Control

In both: NPN Transistor, PNP Transistor, N-Channel/P-Channel (small-signal) MOSFET, TIP120 (NPN Darlington), Relay SPDT, Relay DPDT, Voltage Regulator (5V/3.3V), H-bridge Motor Driver

Tinkercad has, you don't:

- Explicit **3.3V Regulator** as a separately named part (you have one generic "Voltage Regulator")
- Tinkercad explicitly splits **small-signal** vs. **power** MOSFETs into separate named parts; you have one generic N-Channel/P-Channel MOSFET plus a separate power part

You have, Tinkercad doesn't:

- Relay Module (5V) — a pre-built driver board, vs. Tinkercad's bare relay
 - PNP Darlington [TIP125] (Tinkercad only has the NPN TIP120)
 - Power MOSFET [IRF520] as a named part
 - Motor Driver [L293D] named specifically (Tinkercad only offers a generic H-bridge)
 - Optocoupler [4N35] (*exists in Tinkercad too, but under Integrated Circuits — see §3*)
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2.10 Connectors

In both: 8-pin Header, USB Connector

Tinkercad has, you don't: (*none beyond the above two*)

You have, Tinkercad doesn't:

- 8-pin Header (female) — Tinkercad only shows a male header
 - 40-pin Header
 - Screw Terminal Block
 - Barrel Jack
 - Alligator Clip Lead
 - Jumper Wire
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2.11 Logic

This section has the biggest **philosophical** difference between the two tools:

- **Tinkercad's Logic section = physical 74xx chips only** (DIP-package parts with real pinouts): 74HC00, 74HC02, 74HC08, 74HC32, 74HC86, 74HC04, 74HC14, 74HC132, Triple 3-input NAND/AND/NOR (74HC10/11/27), Dual 4-input NAND/AND (74HC20/21), 74HC73/76 (JK/D flip-flops), 74HC93 (4-bit counter), 4-Bit Latch, 4-Bit Adder, 8-Bit Shift Register, Johnson Decade Counter, 7-Segment Decoder, 8-port I²C expander.
- **Your Logic section = mostly ideal/abstract gate symbols** (no chip package, just the logic function), plus one literal chip (Quad NAND [74HC00]).

In both (conceptually): Quad NAND (74HC00 ↔ your "Quad NAND [74HC00]"), the AND/OR/NAND/NOR/XOR functions (ideal in yours, physical chip in Tinkercad's), D and JK flip-flops, 4-bit adder, 4-bit counter.

Tinkercad has, you don't:

- 8-Bit Shift Register (you cap at 4-bit)
- Standalone 4-Bit Latch chip
- 8-port I²C GPIO expander
- Johnson Decade Counter *and* 7-Segment Decoder as **Logic-section** ideal parts (*you have the chip equivalents — CD4017, CD4511 — but they live in your IC section, not Logic*)
- Schmitt-trigger inverters/NAND (74HC14, 74HC132), Triple 3-input gates (74HC10/11/27), Dual 4-input gates (74HC20/21) as **Logic-section** parts (*again, you have these — just as chips under Integrated Circuits*)

You have, Tinkercad doesn't:

- Ideal/schematic-symbol gates as standalone parts: AND, OR, NAND, NOR, XOR, XNOR, NOT, Buffer
 - T Flip-Flop
 - SR Latch
 - Half Adder
 - Multiplexer
 - Demultiplexer
 - 3-to-8 Decoder (ideal symbol — Tinkercad only has it as a chip, and only in the IC-style sense)
 - 4-bit Comparator
 - Clock Generator
 - Logic Toggle (input source)
 - Logic Probe (test/debug tool)
 - Logic Buzzer (test/debug tool)
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2.12 Networking — *section unique to your simulator*

Tinkercad's list you provided has **no networking section or parts at all**.

You have, Tinkercad doesn't:

- Wi-Fi Module [ESP8266]
 - Bluetooth Module [HC-05]
 - Radio Module [nRF24L01]
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3. Cross-Section Differences (same part, filed differently)

These aren't true gaps — the component exists in both tools, just under a different tab:

Component	Tinkercad's section	Your section
IR Remote	Output	Input
Optocoupler [4N35]	Integrated Circuits	Power Control
74HC02 / 04 / 08 / 10 / 11 / 14 / 20 / 21 / 27 / 32 / 86 / 132 / 138 / 4051 (74xx family)	Logic	Integrated Circuits
Decade Counter (CD4017 / Johnson counter)	Logic	Integrated Circuits
7-Segment Decoder (CD4511)	Logic	Integrated Circuits
Potentiometer (generic)	Input (one generic part)	Input, but split into 4 granular parts: Rotary, Trimmer, Slide Potentiometer, Rotary Encoder

Takeaway: your simulator effectively duplicates the 74xx logic family — once as an *ideal gate symbol* (Logic section) and once as a *pinned physical chip* (Integrated Circuits section). Tinkercad only models it once, as the physical chip, filed under Logic. Neither approach is "wrong" — yours gives users both a teaching-friendly ideal gate and a wire-accurate chip; Tinkercad keeps it to one canonical part per chip.

4. Overall Unique-to-Tinkercad Components

(present nowhere in your library, regardless of section) **missing components from mine**

- Ambient Light Sensor
- Generic IR (proximity) Sensor
- SW 2000 (*unidentified — verify*)
- DIP Switch DPST / SPST×4 / SPST×6 variants
- DC Motor with encoder
- 7-Segment Clock Display
- Fixed-size NeoPixel Ring/Strip presets (12/16-LED ring, 6/8/10/16/20-LED strip)
- 556 Dual Timer
- Explicit 3.3V Regulator part, and split small-signal vs. power MOSFET parts
- Breadboard Small, Breadboard Mini
- micro:bit with Breakout
- 8-Bit Shift Register (chip)
- Standalone 4-Bit Latch (chip)
- 8-port I²C GPIO expander

5. Overall Unique-to-Your-Simulator Components

(present nowhere in Tinkercad's list, regardless of section)- **Present in mine absent from thinkersCard**

- **General:** Schottky Diode, Fuse, Crystal Oscillator, Trimmer Capacitor
 - **Input:** Water Level Sensor, Sound Sensor, Hall Effect Sensor, IR Receiver, Real-Time Clock [DS1307], Thermistor (NTC), Vibration Sensor, Magnetic Reed Switch, Phototransistor, Photo Interrupter, Toggle Switch, Slide Potentiometer, Trimmer Potentiometer, Rotary Encoder, Joystick, Limit Switch, Pushbutton (12mm)
 - **Output:** LED (10mm), LED Bar Graph, LCD 20×4, LED Matrix 8×8, 4-Digit 7-Segment Display, Bi-Colour LED, Infrared LED, Continuous Rotation Servo, Stepper Motor, Solenoid, Speaker
 - **Power:** VCC Supply, Ground, 1.5V Battery (AAA), Battery Pack (4×AA)
 - **Microcontrollers:** Nano (ATmega328P)
 - **Integrated Circuits:** Shift Register [74HC595]/[74HC165], Decade Counter [CD4017], BCD → 7-Segment [CD4511], Darlington Array [ULN2003], ADC [MCP3008], EEPROM [24LC256]
 - **Power Control:** Relay Module (5V, pre-built board), PNP Darlington [TIP125], Power MOSFET [IRF520], Motor Driver [L293D] (named)
 - **Connectors:** 8-pin Header (female), 40-pin Header, Screw Terminal Block, Barrel Jack, Alligator Clip Lead, Jumper Wire
 - **Logic:** Ideal gate symbols (AND/OR/NAND/NOR/XOR/XNOR/NOT/Buffer), T Flip-Flop, SR Latch, Half Adder, Multiplexer, Demultiplexer, ideal 3-to-8 Decoder, 4-bit Comparator, Clock Generator, Logic Toggle, Logic Probe, Logic Buzzer
 - **Networking (entire section):** Wi-Fi Module [ESP8266], Bluetooth Module [HC-05], Radio Module [nRF24L01]
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6. Bottom Line

- **Breadth:** Your library is noticeably larger overall (~165 vs. ~104 parts), mainly because of much deeper **Input sensors**, **Connectors**, **Power Control**, and a **dedicated Networking section** Tinkercad simply doesn't have.
- **Where Tinkercad still leads:** size/package variants (breadboard sizes, DIP switch pin-counts, NeoPixel presets, small-signal vs. power MOSFETs), a couple of "convenience" ICs (556 dual timer, 8-bit shift register, I²C expander), and the DC-motor-with-encoder.
- **Biggest structural difference:** how each tool treats the 74xx logic family — Tinkercad keeps one physical-chip version per part under Logic; you maintain both an ideal-gate version (Logic) and a physical-chip version (Integrated Circuits) of the same functions, which is more flexible for teaching but does mean the same chip technically "exists twice" in your library.